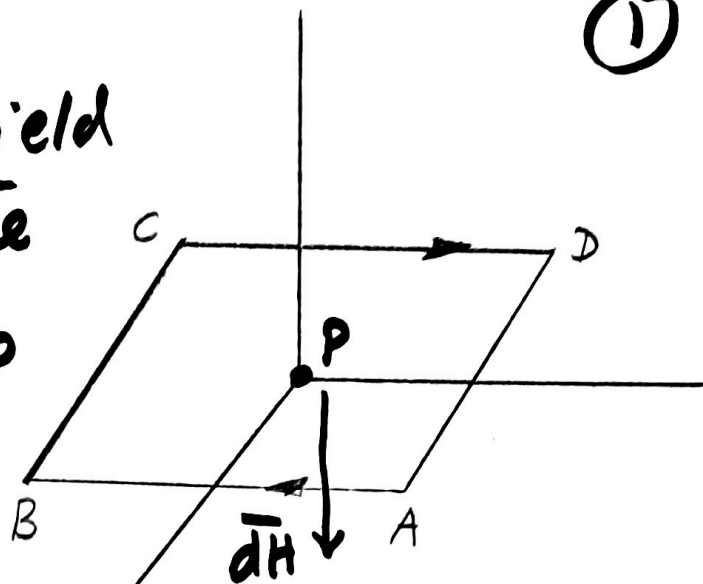


①

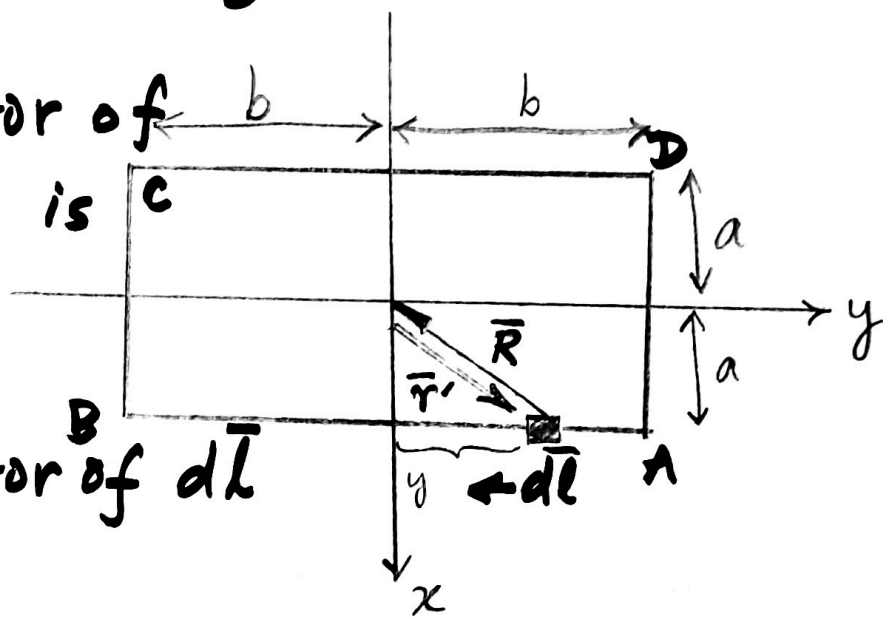
Consider the field point is at the origin, due to

clockwise direction of

current, the field $\vec{dH}$ is into the $x-y$ plane in downward z -axis direction.



The position vector of the field point is $\vec{r} = 0$.



The position vector of $d\vec{l}$ is $\vec{r}'$.

The displacement vector is $\vec{R} = \vec{r} - \vec{r}' = -\vec{r}'$

Note $d\vec{l} = dy \hat{a}_y$.

$$\vec{r}' = a \hat{a}_x + y \hat{a}_y$$

The field $\vec{dH}$ is

$$\vec{dH} = \frac{\mu_0 I}{4\pi} \frac{d\vec{l} \times (\vec{r} - \vec{r}')}{\|\vec{r} - \vec{r}'\|^3}$$

$$\begin{aligned}
 \textcircled{2} \quad d\vec{l} \times (\vec{r} - \vec{r}') &= dy \hat{a}_y (0 - a \hat{a}_x - y \hat{a}_y) \\
 &= -a dy \hat{a}_y \times \hat{a}_x \\
 &\quad - y dy \underbrace{\hat{a}_y \times \hat{a}_y}_{=0} \\
 &= -a dy (-\hat{a}_z) \\
 &= a dy \hat{a}_z
 \end{aligned}$$

$$\begin{aligned}
 \|\vec{r} - \vec{r}'\|^3 &= \|\vec{r}'\|^3 = \|a \hat{a}_x + y \hat{a}_y\|^3 \\
 &= (a^2 + y^2)^{3/2}
 \end{aligned}$$

from point A to B.

$$d\vec{H} = \frac{I}{4\pi} \frac{a dy \hat{a}_z}{(a^2 + y^2)^{3/2}}$$

$$\begin{aligned}
 \vec{H}_{AB} &= \int_{y=-b}^{+b} \frac{I}{4\pi} \frac{a dy \hat{a}_z}{(a^2 + y^2)^{3/2}} \\
 &= \frac{I}{4\pi} a \hat{a}_z \left. \frac{y}{a^2 \sqrt{a^2 + y^2}} \right|_{y=-b}^{+b}
 \end{aligned}$$

$$\bar{H}_{AB} = \frac{I \partial z}{4\pi a} \left(\frac{-b}{\sqrt{a^2 + b^2}} - \frac{b}{\sqrt{a^2 + b^2}} \right) \quad (3)$$

$$\bar{H}_{AB} = \frac{-I b \partial z}{2\pi a \sqrt{a^2 + b^2}} \quad \left(\text{This is down ward in direction as described in the beginning.} \right)$$

The same intensity will be computed when $\bar{dl}$ goes from point C to D.

$$\bar{H}_{CD} = \bar{H}_{AB}.$$

Now consider going from point D to A.

$$\bar{r}' = x \hat{a}_x + b \hat{a}_y$$

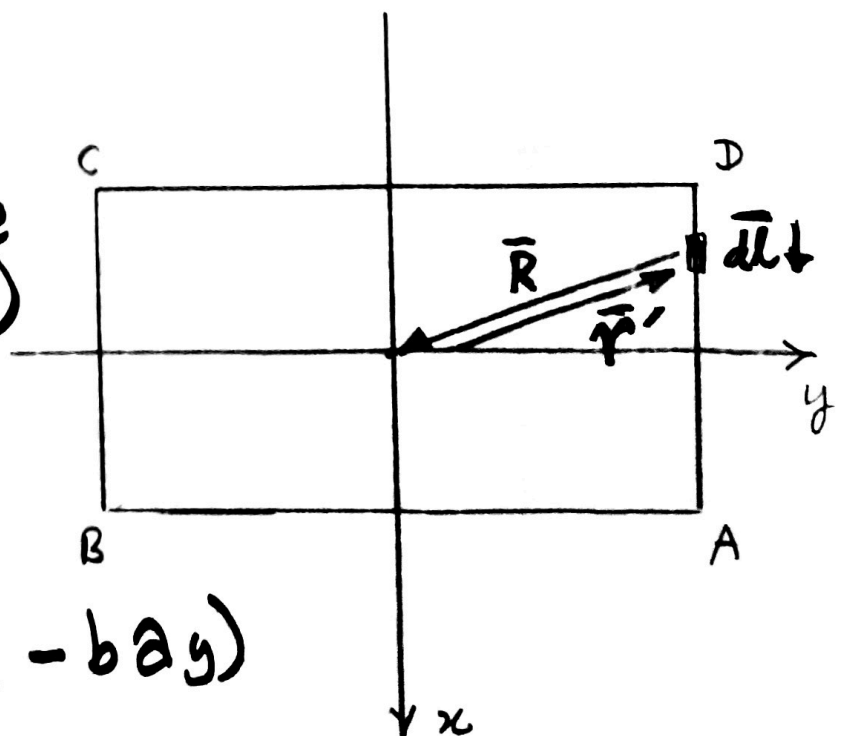
$$\bar{R} = -\bar{r}' \quad (\text{because } \bar{r} = 0)$$

$$\bar{dl} = dx \hat{a}_x$$

$$\bar{dl} \times (\bar{r} - \bar{r}')$$

$$= dx \hat{a}_x \times (-x \hat{a}_x - b \hat{a}_y)$$

$$\bar{dl} = -b dx \hat{a}_z$$



$$\bar{H}_{DA} = \int_{x=-a}^{+a} -\frac{I}{4\pi} \frac{b dx \partial_z}{(x^2 + b^2)^{3/2}} \quad (4)$$

$$= -\frac{I}{4\pi} b \partial_z \int_{x=-a}^{+a} \frac{dx}{(x^2 + b^2)^{3/2}}$$

$$= -\frac{I b \partial_z}{4\pi} \left. \frac{x}{b^2 (x^2 + b^2)^{1/2}} \right|_{-a}^a$$

$$= -\frac{I \partial_z}{4\pi b} \left(\frac{a}{\sqrt{a^2 + b^2}} - \frac{-a}{\sqrt{a^2 + b^2}} \right)$$

$$\bar{H}_{DA} = -\frac{I \partial_z a}{2\pi b \sqrt{a^2 + b^2}}$$

The same result will be obtained for the path B to C.

$$\bar{H}_{BC} = \bar{H}_{DA}.$$

Total Magnetic Field Intensity

⑤

$$\bar{H} = \bar{H}_{AB} + \bar{H}_{BC} + \bar{H}_{CD} + \bar{H}_{DA}$$

$$= - \frac{Ia \Delta z}{\pi b \sqrt{b^2 + a^2}} - \frac{Ib \Delta z}{\pi a \sqrt{b^2 + a^2}}$$

$$\bar{H} = - \frac{I \Delta z}{\pi \sqrt{b^2 + a^2}} \left(\frac{a}{b} + \frac{b}{a} \right)$$